

Structured Research Problem Statement

Project Name:

Optimizing RAG Document Retrieval for Agronomic Advice

Final Values-Led Problem Statement (100–150 words)

Smallholder farmers in Nigeria often struggle to access accurate and timely agricultural information because valuable knowledge is dispersed across extension manuals, research publications, and government resources that are difficult to search effectively. This challenge limits informed decisions on crop diseases, pest management, soil fertility, fertilizer application, and climate adaptation, reducing productivity and income. This project aims to improve access to relevant agricultural information by developing an AI-powered document retrieval system that identifies the most relevant extension documents in response to farmers' questions. The solution will be guided by the values of accessibility, fairness, accuracy, transparency, inclusivity, reliability, and sustainability. By helping farmers quickly locate trustworthy agricultural information, the project seeks to improve farming decisions, strengthen food security, support sustainable agriculture, and enhance the effectiveness of agricultural advisory services.

Section 1: Background and Context

Smallholder farmers, agricultural advisers, and extension workers require accurate and timely information on crop diseases, pests, soil fertility, fertilizers, irrigation, and climate adaptation. Most of this knowledge is stored in extension manuals, research publications, and government advisory documents. Quickly locating the correct documents is essential for informed farming decisions and improved productivity.

Section 2: Core Problem

Farmers often describe their problems using everyday language, while agricultural documents use technical terminology. Because of this mismatch, retrieval systems may fail to identify the most relevant documents and instead return results with similar keywords but limited relevance. Poor retrieval reduces the quality and reliability of RAG-generated advice.

Section 3: Current Approach and Limitations

The baseline approach uses TF-IDF retrieval. Although effective for exact keyword matching, it struggles with synonyms, conversational questions, and differences in terminology. It may also fail to rank the most relevant documents within the top five results.

Section 4: Research Aim

This research aims to develop and evaluate a document retrieval approach that improves the ranking of relevant agricultural extension documents for smallholder farmers' questions. BM25 and hybrid retrieval methods will be compared against the TF-IDF baseline with the goal of improving retrieval quality measured by nDCG@5.

Section 5: Proposed Technical Approach

The project will implement a hybrid retrieval pipeline combining BM25 with sentence-transformer embeddings. A cross-encoder reranker will be investigated to further improve ranking accuracy while maintaining a realistic implementation scope.

Section 6: Scope and Boundaries

The project focuses on a collection of agricultural extension documents covering crops, pests, diseases, soil management, fertilizers, and climate adaptation. It evaluates document retrieval methods only and does not generate final agronomic advice.

Section 7: Evaluation Method

Performance will be evaluated using nDCG@5 and compared against the TF-IDF baseline to determine improvements in ranking relevant documents.

Section 8: Significance

Improving document retrieval will increase the relevance of agricultural information provided to farmers, improve downstream RAG responses, reduce misleading recommendations, and support better agricultural decision-making, productivity, and food security.